



Ontology-based framework for building environmental monitoring and compliance checking under BIM environment

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ABSTRACT

Building environmental monitoring and compliance checking are important in ensuring environmental performance. The information required for monitoring and checking is obtained from different data sources in different information systems. In this context, information sharing between stakeholders and the semantic interoperability that prevails with varying information systems are necessary. However, the implementation of information sharing and semantic interoperability can be a challenge. This paper proposes an ontology-based framework to support environmental monitoring and compliance checking under building information modeling (BIM) environment among different information systems. The framework integrates building information from BIM, environmental information provided by sensors, and regulatory information based on building regulations and design requirements. In this framework, four specific ontologies are developed to represent relevant knowledge. Building information is extracted from BIM and then converted, together with environmental information provided by sensors, into resource description framework format as ontology instances. The regulation clauses are transformed into SPARQL (SPARQL Protocol and RDF Query Language) rules. A case study is performed to apply the framework, and environmental monitoring and automated compliance checking are implemented in the context of a real distributed energy station project. The testing results validate the feasibility and effectiveness of the proposed framework.

1. Introduction

Building environment in this paper refers to the environment of the interior and exterior spaces of building, including indoor air environment, indoor thermal environment, indoor humidity environment, building sound environment, and building light environment. The building environmental performance can significantly affect the health and well-being of people. Therefore, it is significant to monitor and check the building environment (indoor and outdoor environment) during the whole project lifecycle to ensure the building environmental performance. In the pipeline construction phase, the indoor air quality monitoring would be an effective method to ensure a safety-working environment because there may be dangerous or toxic gases at the construction site. During the facility operation phase, the building environmental information are collected in real time for environmental monitoring and assessment. However, due to a lack of unified standards, the information provided by various sensors were in different formats and exhibit obvious heterogeneity, which results in data sharing and interoperability problems [1].

Building environmental monitoring and checking involve public authorities and private institutions (different stakeholders, such as occupants, owners, contractors, and environmental supervision departments) during the whole life cycle of buildings. Building environmental information needs to be shared and exchanged among different stakeholders for different purposes. For example, occupants and owners have concerns about indoor air and moisture. Contractors need to collect information about the working environment, especially in some specific working spaces, to ensure its suitability for workers during the construction stage. Personnel from environmental supervision departments need to check whether the environment is compliant with regulatory requirements.

Different parties generally develop their information systems for different purposes. In this process, separate class hierarchies are used for the same set of objects. That is to say, information is heterogeneous and is represented in different data formats in different information systems. Therefore, interoperation is a challenge for these systems. In addition, information systems are usually developed using object-oriented domain models, thereby leading to several limitations, e.g., (1)

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the domain schema is local and cannot be shared among applications or on the internet, and (2) application schemas cannot be dynamically modified. In this context, information sharing among different stakeholders and semantic interoperability among different information systems become key issues [2].

The knowledge sharing and semantic interoperability is therefore a literature gap for the building environment management, which is the research starting point of this paper. In this work, an ontology-based framework for building environmental monitoring and compliance checking is introduced with the focus on knowledge sharing and interoperability among different information systems via the integration of BIM and sensors. In the proposed framework, the required information is described as ontology instances in resource description framework (RDF) format instead of heterogeneous data formats from different systems. The clauses of building regulations are represented as rules, and then the rules are used to explain the target instances. This framework enables the relevant heterogeneous information for building environmental monitoring and compliance checking to be semantically represented and inferred for rule-based reasoning.

The rest of the paper is organized as follows. In section 2 a brief literature review is given and the research gap is identified. In section 3, the required information and the information semantic fusion model are proposed. The ontology-based framework is also introduced in detail. The framework includes specific ontologies and extraction, conversion, and inference modules. Section 4 presents a real case to validate the feasibility and effectiveness of the proposed framework. Section 5 summarizes the conclusions, limitations, and future research.

2. Related work

This study focuses on building environmental monitoring and compliance checking under BIM environment. Previous research of relevant aspects is reviewed in the following sections.

2.1. Combination of BIM and sensor data

BIM provides a framework to develop data-rich product models. Many information dimensions can be integrated into 3D virtual models and even nD models [3–7,53]. BIM also provides a framework that fosters the integration of information from the conception to the decommissioning of constructed facilities [6]. In recent years, significant work has been done on BIM-based construction facility management and applications [8–10]. BIM-based building environmental monitoring serves an important function in building facility management. Hence, integrating environmental information into BIM has a promising future.

Sensors can be utilized in buildings to collect real-time information, such as temperature, light, and humidity. Many modern buildings are equipped with various monitoring and control systems, e.g., building management systems (BMSs). The majority of BMSs are used to monitor environmental conditions and energy consumption during the operation phase of a building's life cycle.

Arslan et al. developed a prototype system called “real-time environmental monitoring, visualization, and notification system” using BIM and wireless sensor network (WSN) [11]. Riaz et al. proposed a BIM- and sensor-based data management system for automating the management of health and safety issues in construction sites [12]. Natephra et al. proposed the integration of thermal and environmental data provided by sensors with BIM to assess the thermal performance of the building envelope [13]. Suprabhas and Dib discussed the feasibility of using sensor data combined with BIM for maintenance based-facility management [14].

BIM has been used to support applications related to environmental monitoring in certain fields, such as energy performance analysis, CO₂ emissions prediction, and lighting analyses, etc. However, no research has been carried out to propose an integrated application to address the environmental monitoring problem. A main challenge for this is the

lack of common vocabularies among these applications of different fields, which results in an inefficient data interaction. Generally, data derived from one application cannot be shared with others, while the integrated data is critical to be leveraged to solve the environmental monitoring problem. Industry Foundation Classes (IFC) is an open BIM data schema which aims at solving the interoperability problem in data exchange between different systems. However, it is developed mainly for the purpose of building design. Although a new entity IFCSensor has been added to the latest release of IFC, it still needs to be tailored for building environmental monitoring by adding related attributes.

2.2. Environmental monitoring and ontology

Ontology is defined as an explicit and shared conceptualization of a given domain that provides explicit logical assertions about three types of things, classes, instances, and properties, and offers the means to capture and convert human knowledge into a computer-understandable and explicit format [15,16]. In recent years, ontology and semantic technology have been widely used in the construction industry [2,17–27].

Sørensen et al. reviewed existing ontologies relevant to creating digital links between virtual models and physical components in the construction process to improve the information handling and sharing in construction and building operation management [17]. Corry et al. proposed a semantic-based approach to integrating heterogeneous building data [18]. Pauwels et al. investigated the feasibility of semantic web technologies for improving interoperability in the architecture, engineering, and construction industries as an alternative to the existing BIM and Industry Foundation Classes (IFC) approaches [19]. Karan et al. used semantic web technology to ensure semantic interoperability between existing BIM and GIS tools [20]. Mignard and Nicolle proposed an ontology-based approach to solving the heterogeneity problem between BIM and GIS for urban facility management [21].

Semantic web technologies have been used in environmental monitoring to facilitate knowledge encoding, data integration, outside the construction environment [22]. Metal et al. used an ontology to integrate air quality and 3D city models [23]. Opera described an ontology for air pollution analysis and control and presented its use in expert and multi-agent systems [24]. Reitzes and Snyder developed an ontology for real-world indoor environmental quality monitoring and control [25]. Stocker et al. devised an ontology-based environmental monitoring system to measure and compute mean hourly PM_{2.5} concentrations [26]. Pundt et al. described the use of ontologies via the internet on the basis of an example from field GIS-supported environmental monitoring [27]. Dibley et al. proposed an ontology framework for intelligent sensor-based building monitoring with a focus on the ontology development process to deliver an intelligent multi-agent software framework that supports real-time building monitoring [2].

Despite the efforts mentioned above, limited research has been observed in the area of building environment in the construction management domain. The existing efforts mainly focus on the environment at the urban level, and the information related to the building environmental monitoring is not organized effectively. In particular, for the building environment compliance checking, the real time sensing data should be associated with other semantic information, such as the area of the environment, and the type of the room. The literature review highlights the limited semantic modeling of the construction environment at the building level, although BIM, regulation and intelligent sensing models are relevant to this domain. An integrated modeling approach would be beneficial to promote the effective interoperation of software.

2.3. Automated compliance checking

Rule-based compliance checking in construction has received much attention from the academic community and industry over the years

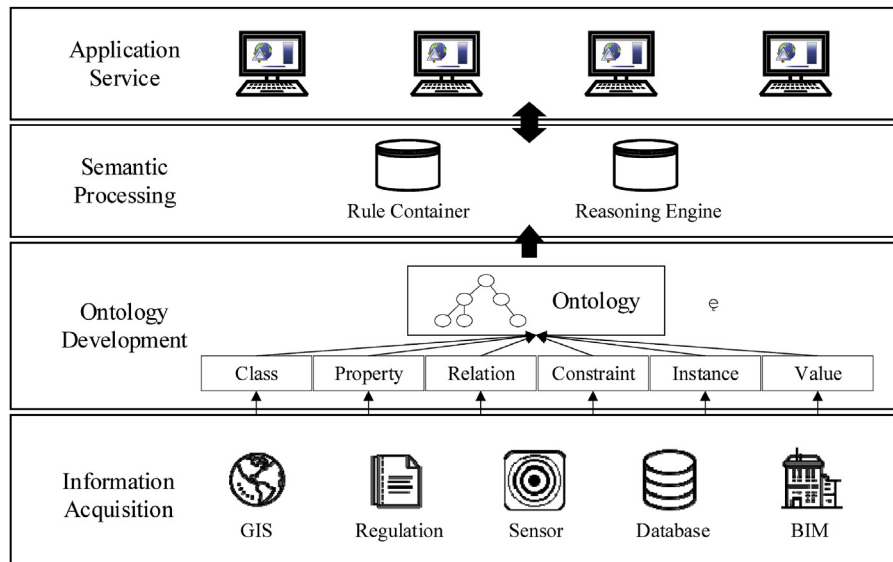


Fig. 1. Ontology-based information semantic fusion model (adapted from Ming et al. [37]).

[28–36]. Solihin and Eastman study and classify the rules for automated BIM rule checking [28]. Fenves proposed a decision table approach for rule-based compliance checking [29]. Nawari used the SmartCode tagging mechanism and Language Integrated Query to compare rules and models [30]. Choi took a similar approach and used SmartCode tagging and XML derived from STEP P21 [31]. Pauwels et al. used N3Logic to create rules from acoustic performance regulations [32]. Zhong et al. used the semantic Web Ontology Language (OWL) and Semantic Web Rule Language technique to create rules from construction quality regulations and then applied the rules into the JESS rule engine [33]. Lee et al. took a similar approach and used OWL and the SPARQL Protocol [34]. Pauwels et al. studied a performance benchmark over semantic rule checking approaches in construction industry [35]. İlal and Günaydın proposed a new hybrid model and methodology to represent building codes for supporting the automated compliance checking [36].

For the environmental monitoring, the reasoning is necessary based on the regulation requirement. The automatic compliance checking of building environment is a particularly important function. Ontology and semantic web technology provide more powerful semantic representation, and enable the data to be interpretable semantically by computers. The semantic inference and retrieval can be easily implemented. However, little efforts related to the building environment compliance are observed.

3. Methodology for building environmental monitoring and compliance checking

3.1. Information requirement

Building environmental monitoring and compliance checking mainly require information about building elements, environmental data, and regulation clauses.

- (1) Building environmental data, including temperature, humidity, noise, and air quality, etc., are usually collected using various sensors in real time.
- (2) Building regulation clauses usually stipulate the threshold of particular parameters in certain circumstances (e.g., concentration limit of partial environmental pollutants in building interior environment).
- (3) Building information includes the geometry and alphanumeric data of building elements and various types of building demand. What

information that should be extracted from BIM depends on what type of sensor and regulation requirements. For example, for the indoor air quality monitoring, the sensor numbers, the location of sensors, geographical information about the room, etc., are required.

As mentioned in Section 1, the information above is stored and represented in different data formats from different information systems. This condition presents a significant challenge in facilitating information and knowledge sharing, as well as the automation of compliance checking. To solve this problem, we design an ontology-based information semantic fusion model.

3.2. Ontology-based information semantic fusion model

The semantic fusion model was designed to support effective information semantic management and service applications by attaching semantic concepts at different abstraction levels of information from different sources and using domain knowledge-based rules for reasoning.

The model consisted of four layers: (1) information acquisition (2) ontology development; (3) semantic processing; and (4) application service. In the information acquisition layer, heterogeneous information provided by different data sources, including sensors, was collected and usually stored in several forms, such as databases, text files, GIS, and BIM. In the ontology development layer, specific ontologies are developed to describe the knowledge and information of different domains. The relationships among the defined concepts are used as the basis of reasoning. In the semantic processing layer, rule container and reasoning engine are created and used to support the management and application of ontology-described information. Finally, in the application service layer, multiple service applications, including environmental monitoring, compliance checking, information storage, and information sharing services, are provided through programmable interfaces for users (see Fig. 1).

3.3. Ontology-based framework for building environmental monitoring and compliance checking

On the basis of the information requirements and semantic fusion model given above, the framework for building environmental monitoring and compliance checking was developed by leveraging state-of-the-art technologies, as shown in Fig. 2.

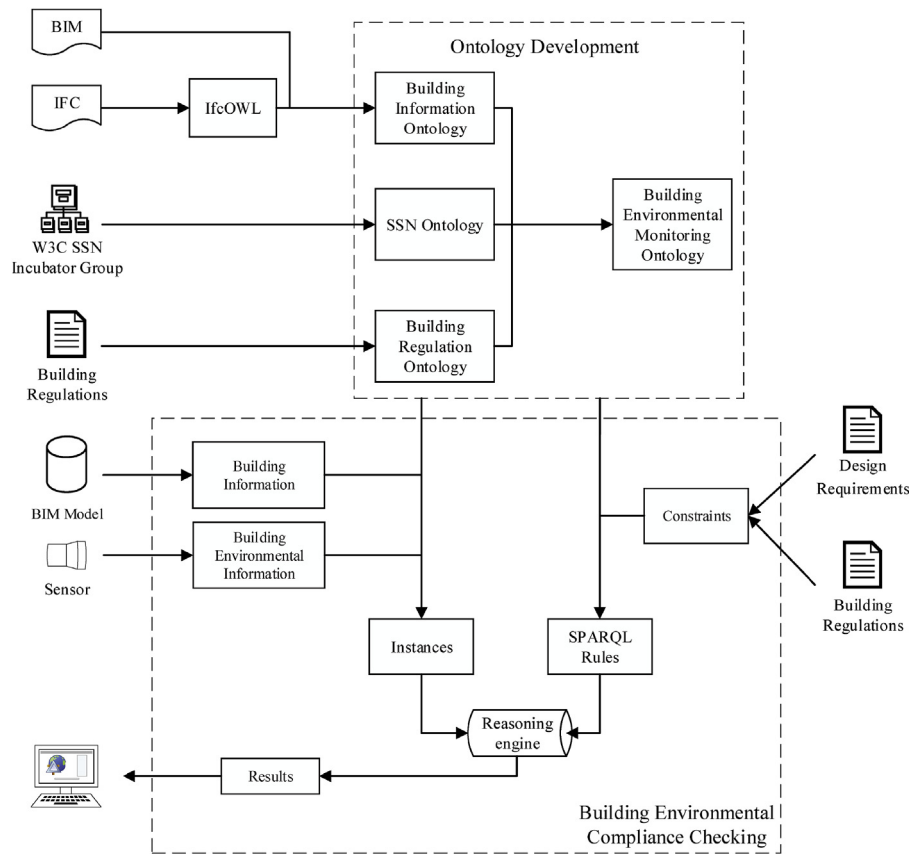


Fig. 2. Proposed framework for building environmental monitoring and compliance checking.

The information related to building environment and the corresponding constraints are two vital elements for building environmental monitoring and compliance checking. Ontologies are used to represent various domain knowledge and integrate different information related to building environment. Building information is retrieved from BIM. Building environmental information is provided by sensors. Building regulations and design requirements are converted to rules and constraints. SPARQL queries are used to formalize the rules and constraints. A reasoning engine is used to infer logical consequences on the basis of rule checking.

As shown in this framework, four ontologies (building information ontology, semantic sensor network (SSN) ontology, building regulation, and building environmental monitoring) act as the backbone. The first three ontologies are developed to represent the knowledge and information of various domains, and the last one is developed such that it is associated with the other three specific ontologies. Domain knowledge is expressed as classes, properties, and relationships in ontologies. BIM model provides the static building information and various sensors provide the dynamic building environmental information. The building information and building environmental information are represented as ontology instances. The building regulatory constraints are modeled into OWL axioms and SPARQL rules. The details of this framework are presented below.

3.3.1. Ontology development

3.3.1.1. Building information ontology. The building information ontology was developed to represent building knowledge and information from BIM (Fig. 3). It is also developed on the basis of the ifcOWL ontology [38].

The ifcOWL ontology [38] is an OWL representation of the IFC and serves as an alternative representation of the EXPRESS schema of the IFC. The IFC allows the representation of building information modeled

in a 3D BIM environment. The creation of the ifcOWL ontology relies on a previous EXPRESS-to-OWL conversion approach [39]. This complex approach maps each element in the EXPRESS file onto its nearest equivalent in OWL, thereby producing a solid and trustworthy ontology.

According to the specific requirements of the underlying problem, the proposed building information ontology simplifies the classes, attributes, and relationships of the ifcOWL ontology and supplements required classes, attributes, and relationships. In the building information ontology, the classes have the prefix “BI” to avoid repetition and ambiguity. These classes mainly include BI-BuildingDemand, BI-BuildingElement, and BI-Parameter, as well as their subclasses. These instances represent actual building information from BIM.

BI-Room refers to the room contained in a building and is the subclass of BI-IfcSpace. BI-CivilBuildingCategory refers to the category of a building and is the subclass of BI-BuildingCategory. BI-IfcSensor refers to the sensor which measures a physical quantity and converts it into a signal. BI-Room is affected by BI-BuildingCategory.

3.3.1.2. SSN ontology. SSN ontology was used to represent sensor knowledge and environmental information provided by sensors (Fig. 4). SSN ontology was developed by the World Wide Web Consortium (W3C) Semantic Sensor Networks Incubator Group, and it describes sensors and observations and related concepts. This ontology does not describe domain concepts, time, and locations. This information can be provided by other ontologies via OWL imports [40]. The location information of sensors is described in building information ontology.

The classes of SSN ontology have the prefix “SSN” and mainly include SSN-Sensor, SSN-SensorOutput, and SSN-ObservationValue. These instances represent building environmental information provided by sensors.

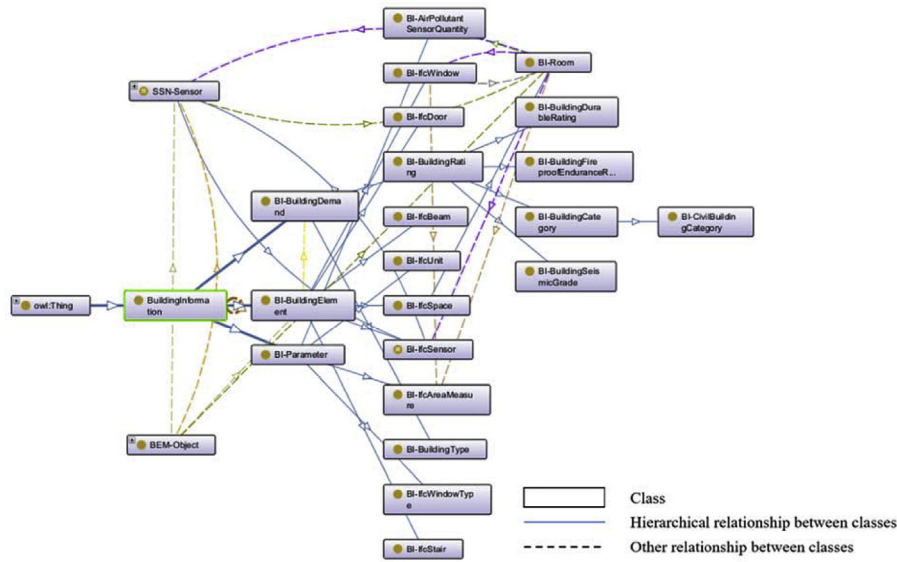


Fig. 3. Building information ontology.

SSN-Sensor refers to the sensor and is equivalent to BI-IcSensor. SSN-SensorOutput refers to the sensor output. SSN-Observation refers to the content of sensor output and SSN-ObservationValue refers to the accurate value of content. These classes are related by object attributes: SSN-Sensor has output SSN-SensorOutput; SSN-SensorOutput has SSN-Observation; and SSN-Observation has SSN-ObservationValue.

3.3.1.3. Building regulation ontology. Building regulation ontology was developed to represent the knowledge of building regulations (Fig. 5). Furthermore, it defines generic concepts and relationships. Then, the constraints in building regulations can be modeled into OWL axioms and SPARQL rules.

Zhong et al. proposed an ontology of construction quality regulation for construction quality inspection and evaluation [33]. Referring to the existing ontology mentioned above, we developed a building regulation ontology and assigned its classes with the prefix “BR.” The classes mainly include BR-Document, BR-Clause, BR-ScenarioContent, and BR-ConstraintContent.

BR-Document refers to the building regulation document and BR-Clause refers to the clause in building regulation document. A clause is usually divided into two parts: the content of the scenario and the content of the constraint. BR-ScenarioContent and BR-ConstraintContent refer to the content of the scenario and the content of the constraint respectively. BR-Clause belongs to BR-Document, and has BR-ScenarioContent and BR-ConstraintContent.

3.3.1.4. Building environmental monitoring ontology. As shown in Fig. 6, the building environmental monitoring ontology defines the concepts and relationships of the building environmental monitoring domain and creates subsequent relationships with other ontologies.

In Ref. [18], an ontology for building performance assessment was developed by Corry et al. as a means to define the performance assessment domain and subsequent relationships with other ontologies.

Similarly, in this framework, the building environmental monitoring ontology is linked to the other three ontologies to acquire and integrate the information of the three ontologies. It also supports the reasoning between rules and ontology instances.

The classes of the building environmental monitoring ontology have the prefix “BEM,” and they mainly include BEM-Object, BEM-Value, and BEM-ConstraintCondition. Some classes are associated with the corresponding classes of the three ontologies.

BEM-Object refers to the managed object and is related to SSN-Sensor. BEM-EnvironmentObjectType refers to the environment type, such as air, temperature and noise, and is related to SSN-ObservationValue.

The ontology was developed using Protégé 5.1 platform, as shown in Fig. 7. First, the related concepts and terms were extracted from the relevant domain documents and then refined and added to form the basic set of concepts and terms with the support of domain experts. Then, the ontology classes were built on the basis of the basic set of concepts and terms. Next, the properties of classes, constraints of

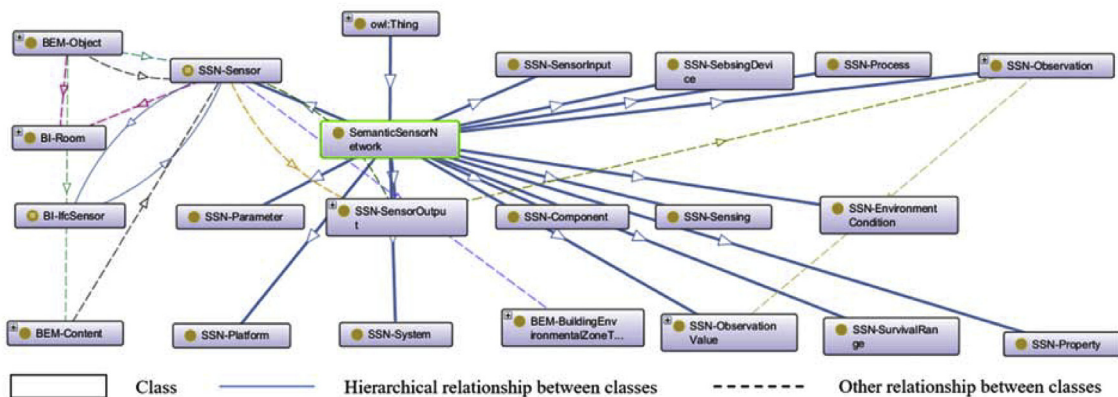


Fig. 4. SSN ontology.

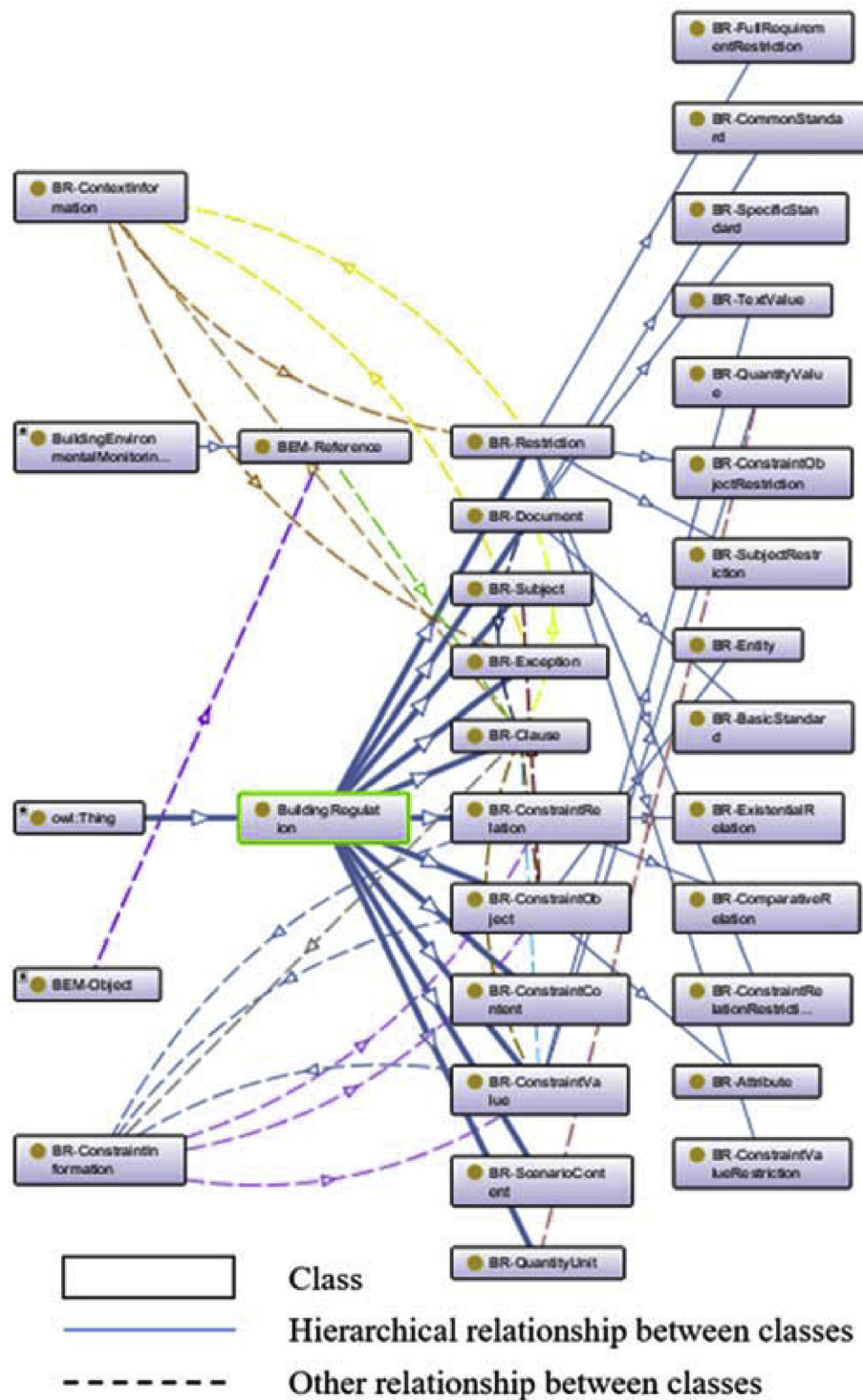


Fig. 5. Building regulation ontology.

objects, and relationships between classes were established on the basis of the analysis and actual conditions. Finally, ontology instances were created to represent real building and environmental information.

3.3.2. Information extracted from BIM

On the basis of ontologies, the required information can be represented as ontology instances and then used for compliance checking.

Here given that almost all building information of the project is integrated into BIM model and can be extracted from it. The required information in BIM must first be extracted and then converted into RDF.

Some research efforts have focused on extracting specific information from BIM model through semantic web technologies and domain-specific ontologies. Zhang et al. used ontology in the extraction of a BIM model from the original complete model [41]. Lee et al. used the embedded software functions of Autodesk Revit to extract context information from BIM model [34]. Nepal et al. proposed the process and approaches to automate the querying of construction-specific information from BIM model [42] and then proposed a novel approach that utilizes ontology-based feature modeling, automatic feature extraction, and query processing to extract information relevant to construction

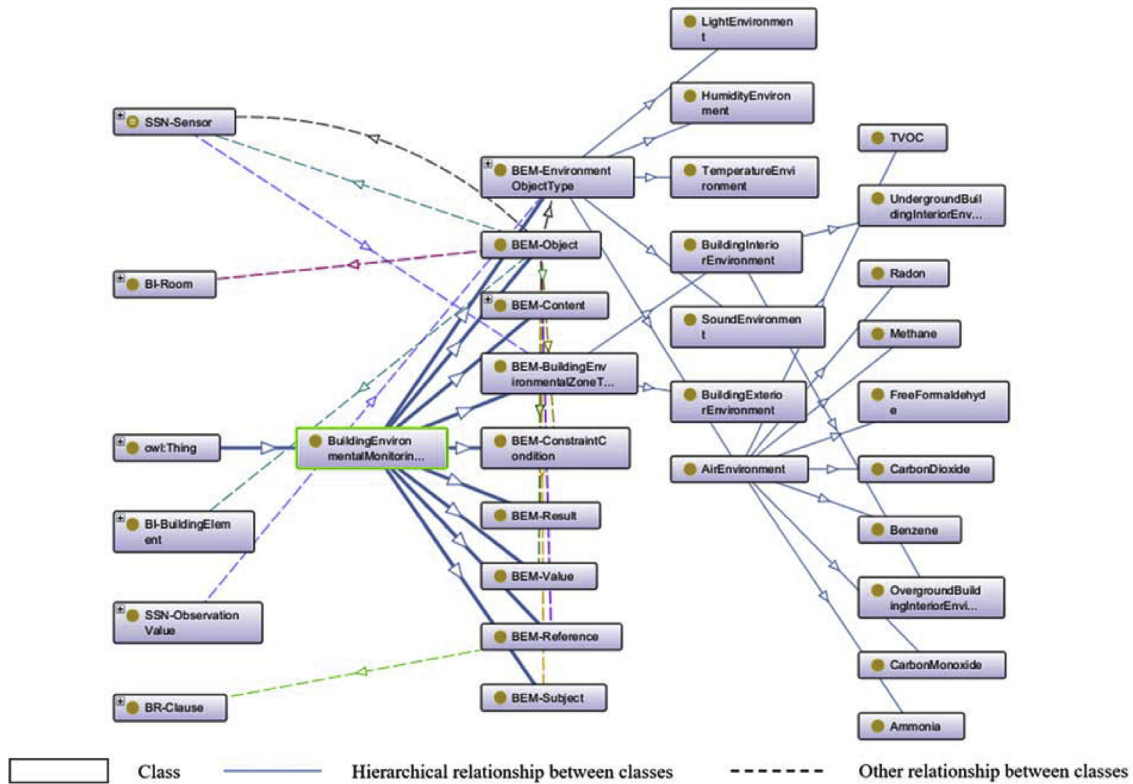


Fig. 6. Building environmental monitoring ontology.

practitioners from BIM model [43]. Demir et al. extracted the instance “exterior wall” from an ifcOWL building model by using SPARQL mapping and query functions [44].

Here, the building information in BIM is extracted using the embedded functions of Autodesk Revit. Initially, parameters such as sensor quantity and building category are created at the project level to represent the required building information. Then, the building information is inputted into the corresponding parameters manually or semi-automatically. Thereafter, the information can be extracted by the detail table function. Finally, the extracted information can be stored in a tabular format and exported to a text file, as shown in Fig. 8. With the embedded functions, users can easily retrieve the required building information and express the information in a structured format. The structured information can be converted from spreadsheet to an RDF/XML file.

3.3.3. RDF conversion

The extracted information needs to be converted from a tabular

format to RDF/XML format as ontology instances. An open-source library called dotNetRDF [45] is utilized to handle the conversion process according to the work of Lee et al. [34]. The customized conversion tool utilizing this library is developed and written in C# language. To use this tool, users need to specify the path of the target spreadsheet. First, the text file of the extracted information needs to be transformed into a spreadsheet. Then, the tool handles the relationship generation among different data items according to the created ontologies mentioned in previous sections. Finally, a file in RDF/XML format is generated as the output. Fig. 9 illustrates the result of the conversion of building information into an RDF/XML file using the developed tool.

3.3.4. Regulatory constraint representation

As mentioned above, the ontology is used to represent domain knowledge and information from BIM models and sensors. Ontology can help enforce constraints on a class level, but the complex constraint knowledge, which typically occurs in the form of a rule, cannot be expressed [46]. Therefore, the expression of complex constraint

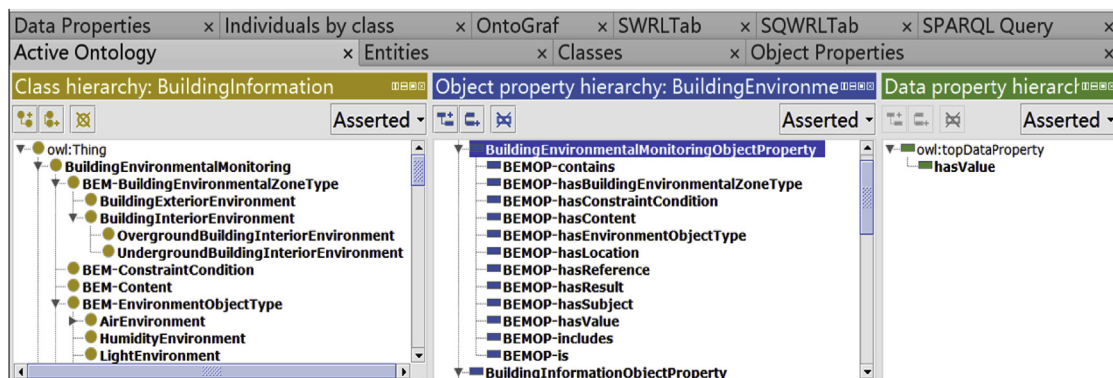


Fig. 7. The building environmental monitoring ontology developed using Protégé.

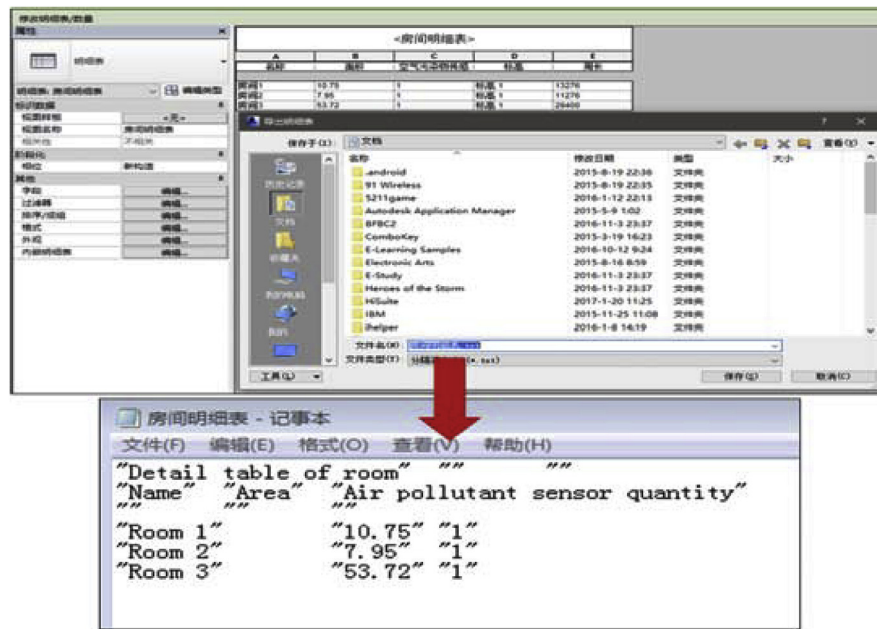


Fig. 8. Information extraction process.

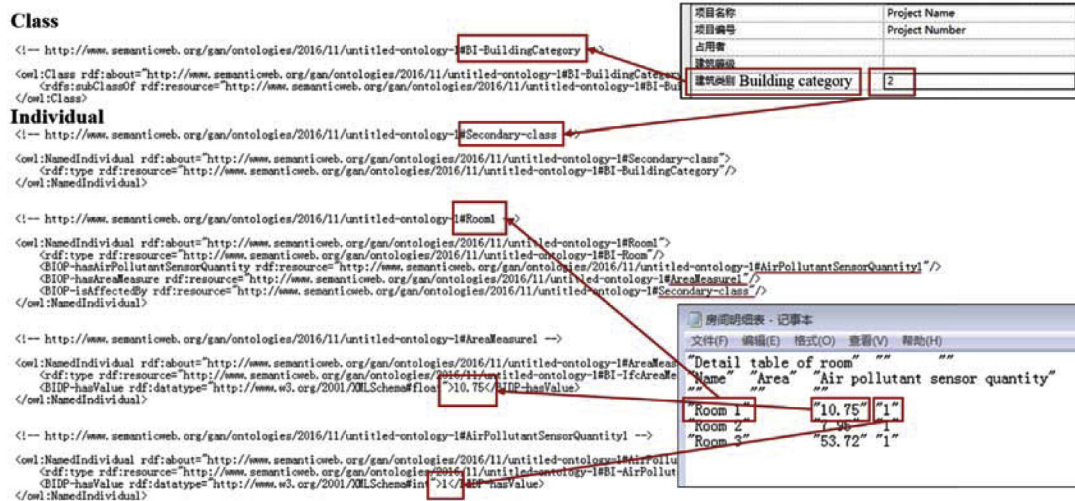


Fig. 9. Result of conversion process.

knowledge requires the implementation of other technologies. SPARQL is a good choice to represent constraint knowledge.

SPARQL, as an RDF query language, can be used to retrieve and manipulate data in RDF [47]. It was developed as a standard by the RDF Data Access Working Group of the W3C and is recognized as one of the key semantic web technologies. This query language allows triple query patterns, conjunctions, disjunctions, and optional patterns. As a standard query language, SPARQL is relatively easy to reuse defined queries or rules in many other platforms and is particularly good in providing extensibility of the rule set [48].

The constraints are represented as SPARQL rules, whereas building information and building environmental information are represented as logic facts. The results of the reasoning show the capability and flexibility to retrieve meaningful information from organized RDF/XML data.

The ontology-based SPARQL rules of regulatory knowledge representation are created in accordance with the following three steps.

- (1) Conversion of building regulation texts into a formal language such

as RDF and OWL. This step has been implemented in section 3.3.1 that the building regulation ontology has been developed.

- (2) Identification of required concepts of the regulatory constraint in the building regulation ontology.
- (3) Creation of SPARQL rules that represent the regulatory constraint in building regulation texts.

On the basis of building regulatory clauses and design requirements, some constraints are obtained and then used to create SPARQL rules. For example, clause 6.0.4 of building regulation GB 50325-2010 (2013) stipulates the concentration limit of radon in the interior environment of secondary-class civil buildings. In addition, the concentration range of radon must be less than or equal to 400 Bq/m³. The corresponding SPARQL rule related to the concentration range of radon is created, as shown in Fragment 1.

These SPARQL rules can be operated using the SPARQL query plugin in the protégé or ARQ (a SPARQL 1.1 compliant engine in the Jena). In this paper, the SPARQL query plugin is used, and the interface of this plugin in the protégé 5.1 is shown in Fig. 10.

Fragment 1

Rule A related to the concentration range of radon in building interior environment.

```
PREFIX ont: < http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1# >
PREFIX rdf: < http://www.w3.org/1999/02/22-rdf-syntax-ns# >
PREFIX owl: < http://www.w3.org/2002/07/owl# >
PREFIX rdfs: < http://www.w3.org/2000/01/rdf-schema# >
PREFIX xsd: < http://www.w3.org/2001/XMLSchema# >
SELECT ?Sensor ?Room ?Radon
WHERE {
  ?Sensor rdf:type ont:SSN-Sensor.
  ?Sensor ont:BEMOP-hasLocation ?Room.
  ?Room rdf:type ont:BI-Room.
  ?SensorOutput ont:SSNOP-isProduceBy ?Sensor.
  ?SensorOutput ont:SSNOP-hasObservation ?Observation.
  ?Observation ont:SSNOP-hasObservationValue ?ObservationValue.
  ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?B.
  ?B rdf:type ont:Radon.
  ?ObservationValue ont:hasValue ?Radon.
  filter (?Radon > 400)
}
```

4. Case study**4.1. Project information**

To validate the feasibility and effectiveness of the proposed framework, we use the monitoring application context of a real distributed energy station project, the International Exhibition Center, as the background of an application case.

The International Exhibition Center is powered by a distributed energy station, which provides full loads of electricity for air conditioning and hot water. The station has many specialized equipment and pipelines arranged in a small space. As a result, the maintenance of the station is a demanding and complex work. For example, it is necessary to monitor the indoor environment to prevent the corrosion of these pipelines, and it is important to ensure safe, healthy and comfortable indoor environments of working place for the site workers and facility operation.

To facilitate the facility management, several types of sensors were deployed in the distributed energy station. A BIM of the station was compiled, and the sensors were incorporated in the model, as shown in Fig. 11. Considering the constraints of related building regulations and user requirements, the major items to monitor include temperature, humidity, noise, light and air quality (such as carbon monoxide, carbon dioxide, methane, radon, free formaldehyde, benzene, ammonia, and total volatile organic compounds (TVOC)) in the distributed energy

station. The information from BIM and sensors is used for building environmental monitoring and compliance checking.

4.2. Regulatory constraint

Implementing building interior environmental monitoring involves four related building regulations: “Code for indoor environmental pollution control of civil building engineering (GB 50325-2010 (2013))”, “Emission standard of environment noise for boundary of construction site (GB 12523-2011)”, “Indoor air quality standard (GB/T 18883-2002)”, and “Standard for lighting design of buildings (GB 50034-2013)”. Regulation GB 50325-2010 (2013) is implemented to prevent and control the indoor environmental pollution caused by building and decoration materials in civil construction projects to ensure public health and benefit [49]. The regulation GB 12523-2011 is implemented to prevent and control the noise pollution of building operations and improve the quality of the acoustic environment [50]. The regulation GB/T 18883-2002 is implemented to protect human health and prevent and control indoor air pollution [51]. The regulation GB 50034-2013 is implemented to promote green lighting applications and satisfy the needs of the building function, which is conducive to production, work, study, life and physical and mental health [52].

These building regulations are mainly composed of three parts: terminology, environment monitoring methods and constraint requirements. The constraint requirements are mainly the quantitative

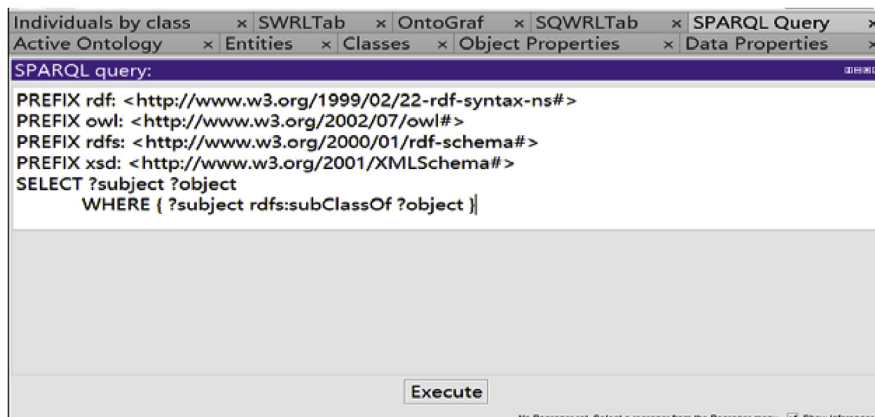


Fig. 10. Interface of SPARQL query plugin.

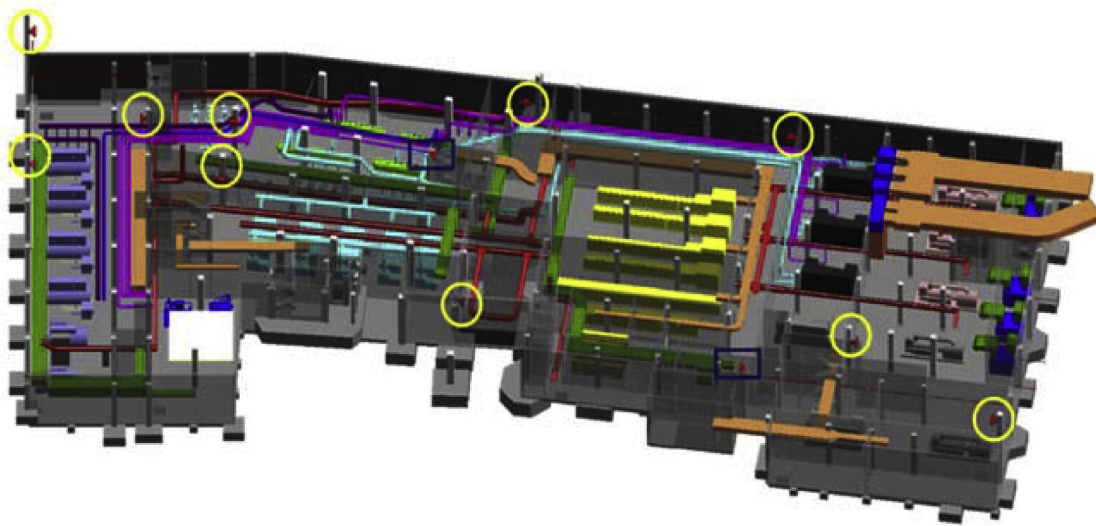


Fig. 11. The partial BIM model of the distributed energy station of the International Exhibition Center.

requirements about the specific building environmental indicators.

For example, the clause 6.0.4 of building regulation GB 50325-2010 (2013) stipulates the concentration constraints of partial air pollutants in building interior environment, including radon, free formaldehyde, benzene, ammonia, and TVOC, the clause 4.1 of building regulation GB 12523-2011 stipulates the maximum value of noise, and the clause 5.3 of building regulation GB 50034-2013 stipulates the minimum value of light in the public building. The user requirements about temperature and humidity in the distributed energy station are also considered. Some constraints related to building interior environmental monitoring are listed in Table 1.

Table 2 lists the number of air pollutant sensors required for the building interior space of different sizes. This requirement is obtained from clause 6.0.14 of building regulation GB 50325-2010 (2013), which stipulates the correlation between the quantity of air pollutant sensors and room area.

These constraints can be converted into SPARQL rules for building environmental monitoring and compliance checking. SPARQL rules are shown in Table 3.

4.3. Building information and sensing data

As mentioned above, information for compliance checking is acquired from different data sources. The semantic interoperability between the building and the environmental information needs to be addressed [2]. Therefore, building information is extracted from the BIM of this project, and the real building environmental information collected by sensors is converted into RDF format, on the basis of the framework given in Section 3.

Here, take the indoor air monitoring for example, the room name,

Table 2

Constraints about air pollutant sensor quantity.

Room area (m ²)	Number of air pollutant sensors
< 50	1
≥ 50, < 100	2
≥ 100, < 500	≥ 3
≥ 500, < 1000	≥ 5
≥ 1000, < 3000	≥ 6
≥ 3000	≥ 3/1000m ²

room area, and quantity of air pollutant sensors, etc. are extracted from BIM (Table 4).

The required building environmental information provided by sensors includes temperature, humidity, noise, and air quality, etc. (Table 5). The data are collected via different sensors (such as visible light sensors, carbon dioxide sensors, temperature sensors, etc.), and then transformed via WIFI and stored in .CSV in the cloud server.

On the basis of the proposed framework, the information from the BIM and sensors is converted into ontology instances and stored in RDF format. The RDF graph of ontology instance is shown in Fig. 12.

4.4. Rule reasoning for compliance checking

The rule reasoning process is shown in Fig. 13. The SPARQL rules are based on the constraint knowledge in building regulations and concepts and relationships of ontology, and are created through the SPARQL generator. The constraint objects of SPARQL rules include the quality of air (such as radon and carbon monoxide), temperature, light, and correlation between the quantity of air pollutant sensors and room

Table 1

Some constraints about building interior environmental monitoring.

Parameter	Temperature	Humidity	Noise	Carbon monoxide (CO)	Carbon dioxide (CO ₂)	Methane (CH ₄)
Unit	°C	%	dB(A)	ppm	ppm	LEL
Normal range	0–38	50–90	≤ 70(day) ≤ 55(night)	< 20	< 8000	< 10
Parameter	Radon (Rn)	Free formaldehyde (HCHO)	Benzene (C ₆ H ₆)	Ammonia (NH ₃)	TVOC	Light
Unit	(Bq/m ³)	(mg/m ³)	(mg/m ³)	(mg/m ³)	(mg/m ³)	lx
Normal range	≤ 400	≤ 0.10	≤ 0.09	≤ 0.2	≤ 0.6	≥ 300

Table 3
SPARQL rules.

Rule No.	Constraint object	SPARQL rule
A	Radon	<pre> SELECT ?Sensor ?Room ?Radon WHERE { ?Sensor rdf:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdf:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdf:type ont:Radon. ?ObservationValue ont:hasValue ?Radon. filter (?Radon > 400) } </pre>
B	Carbon monoxide	<pre> SELECT ?Sensor ?Room ?CarbonMonoxide WHERE { ?Sensor rdf:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdf:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdf:type ont:CarbonMonoxide. ?ObservationValue ont:hasValue ?CarbonMonoxide. filter (?CarbonMonoxide > 20) ?CarbonMonoxide = 20 } </pre>
C	Temperature	<pre> SELECT ?Sensor ?Room ?Temperature WHERE { ?Sensor rdf:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdf:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdf:type ont:TemperatureEnvironment. ?ObservationValue ont:hasValue ?Temperature. filter (?Temperature < 0 ?Temperature > 38) } </pre>
D	Light	<pre> SELECT ?Sensor ?Room ?Light WHERE { ?Sensor rdf:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdf:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdf:type ont:LightEnvironment. ?ObservationValue ont:hasValue ?Light. filter (?Light < 300) } </pre>
E	Correlation between the quantity of air pollutant sensors and room area	<pre> SELECT ?Room ?AreaMeasure ?AirPollutantSensorQuantity WHERE { ?Room rdf:type ont:BI-Room. ?Room ont:BIOP-hasAreaMeasure ?A. ?A ont:BIDP-hasValue ?AreaMeasure. ?Room ont:BIOP-hasAirPollutantSensorQuantity ?B. ?B ont:hasValue ?AirPollutantSensorQuantity. filter((?AreaMeasure < 50 && ?AirPollutantSensorQuantity < 1) (?AreaMeasure >= 50 && ?AreaMeasure < 100 && ?AirPollutantSensorQuantity < 2) (?AreaMeasure >= 100 && ?AreaMeasure < 500 & & ?AirPollutantSensorQuantity < 3) (?AreaMeasure >= 500 && ?AreaMeasure < 1000 && ?AirPollutantSensorQuantity < 5) (?AreaMeasure >= 1000 && ?AreaMeasure < 3000 && ?AirPollutantSensorQuantity < 6) (?AreaMeasure >= 3000 && ?AirPollutantSensorQuantity < 3 * ?AreaMeasure / 1000)) } </pre>
F	...	...

Table 4
Partial collected information on sensor quantity and room area.

Room No.	Room Area (m ²)	Quantity of air pollutant sensors
1	10.75	1
2	7.95	1
3	53.72	1
4	12.00	2
5	85.00	2
6	110.00	2

area. The SPARQL query program is made available on top of the rule reasoning engine, so that the user can use this program to query the results of the rules. The query results are the non-compliant ontology instances.

Rule A is about the concentration range of radon. The query result is listed at the bottom corner of Fig. 14, which shows the values of radon in Room 2 provided by Sensor 3 and Room 3 provided by Sensor 5.

Rule B is about the concentration range of carbon monoxide. The query result is listed at the bottom corner of Fig. 15, which shows the values of carbon monoxide in Room 2 provided by Sensor 3 and Room 3 provided by Sensor 5.

Table 5
Partial collected information on building environment.

Sensor No.	Location	Monitoring type	Temperature	Humidity	CO	CO ₂	CH ₄	Rn	HCOC	C ₆ H ₆	NH ₃	TVOC	Light
1	Room 1	Air	/	/	10	7000	5	100	0.05	0.05	0.1	0.3	/
2	Room 1	Humidity Temperature	30	60	/	/	/	/	/	/	/	/	/
3	Room 2	Air	/	/	30	7000	5	600	0.05	0.05	0.1	0.3	/
4	Room 2	Humidity Temperature	50	91	/	/	/	/	/	/	/	/	/
5	Room 3	Air	/	/	60	7000	5	420	0.05	0.05	0.1	0.3	/
6	Room 3	Humidity Temperature	45	60	/	/	/	/	/	/	/	/	/
7	Room 4	Light	/	/	/	/	/	/	/	/	/	/	300
8	Room 5	Light	/	/	/	/	/	/	/	/	/	/	270
9	Room 6	Light	/	/	/	/	/	/	/	/	/	/	200

Rule C is about the temperature range. The query result is listed at the bottom corner of Fig. 16, which shows the values of temperature in Room 2 provided by Sensor 4 and Room 3 provided by Sensor 6.

Rule D is about the Light range. The query result is listed at the bottom corner of Fig. 17, which shows the values of light in Room 5 provided by Sensor 8 and Room 6 provided by Sensor 9.

Rule E is about the correlation between the quantity of air pollutant sensors and room area. The query result is listed at the bottom corner of Fig. 18, which shows the room area and quantity of air pollutant sensors of Room 3 and Room 6.

4.5. Framework validation

The SPARQL query is used to check the various environmental indicators of each room, and the query results mentioned above show the indicators that do not conform to the specific constraints. The environmental checking results of each room are shown in Table 6, in which, “Pass” means that the environmental indicator meets the requirement of constraint, and “Warning” means that the environmental indicator does not meet the requirement of constraint and need to be focused.

The comparison between the results of automatic reasoning using the proposed approach and manual judgment is used to validate the feasibility of the proposed framework. The objects, that do not meet the requirements of constraints, are reasoned respectively by the automatic reasoning and manual judgment and listed, as shown in Table 7. It can be seen that the results of automatic reasoning are exactly the same as the results of manual judgment.

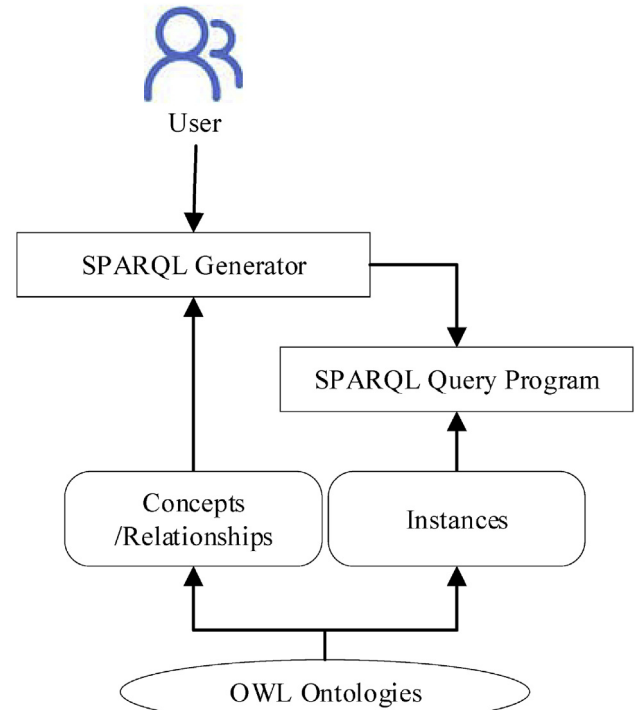


Fig. 13. Rule reasoning process for compliance checking.

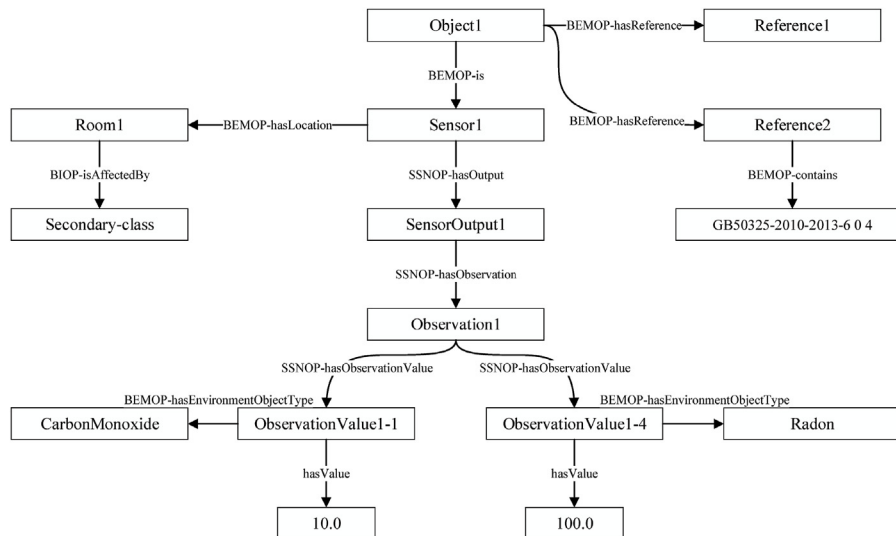


Fig. 12. RDF graph of ontology instance.

SPARQL query:		
<pre> PREFIX ont: <http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1#> PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#> SELECT ?Sensor ?Room ?Radon WHERE { ?Sensor rdfs:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdfs:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdfs:type ont:Radon. ?ObservationValue ont:hasValue ?Radon. filter (?Radon > 400) }</pre>		
Sensor	Room	Radon
Sensor5	Room3	"420"^^<http://www.w3.org/2001/XMLSchema#>
Sensor3	Room2	"600"^^<http://www.w3.org/2001/XMLSchema#>

Fig. 14. Query result of rule A.

SPARQL query:		
<pre> PREFIX ont: <http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1#> PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#> SELECT ?Sensor ?Room ?CarbonMonoxide WHERE { ?Sensor rdfs:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdfs:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?B. ?B rdfs:type ont:CarbonMonoxide. ?ObservationValue ont:hasValue ?CarbonMonoxide. filter (?CarbonMonoxide > 20 ?CarbonMonoxide = 20) }</pre>		
Sensor	Room	CarbonMonoxide
Sensor5	Room3	"60"^^<http://www.w3.org/2001/XMLSchema#>
Sensor3	Room2	"30"^^<http://www.w3.org/2001/XMLSchema#>

Fig. 15. Query result of rule B.

SPARQL query:		
<pre> PREFIX ont: <http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1#> PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#> SELECT ?Sensor ?Room ?Temperature WHERE { ?Sensor rdfs:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdfs:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdfs:type ont:TemperatureEnvironment. ?ObservationValue ont:hasValue ?Temperature. filter (?Temperature < 0 ?Temperature > 38) }</pre>		
Sensor	Room	Temperature
Sensor4	Room2	"50"^^<http://www.w3.org/2001/XMLSchema#>
Sensor6	Room3	"45"^^<http://www.w3.org/2001/XMLSchema#>

Fig. 16. Query result of rule C.

In our framework, the semantic web technologies are used for the building environmental information and corresponding rules. The great advantage of using semantic web technologies is that the schema, the instances, and the rules can all be described using the same format or language. Moreover, ontology and semantic web technology have powerful semantic expression capability, and enable the information to

be interpretable semantically by computer [48].

In addition, the representation capability of the constraint rules and the potential of semantic query depend on the concepts, relationships and instances defined in the ontology. This representation capability can be solved, since the ontology has strong extensibility and operability to improve the concepts, relationships and instances.

SPARQL query:		
<pre> PREFIX ont: <http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1#> PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#> SELECT ?Sensor ?Room ?Light WHERE { ?Sensor rdf:type ont:SSN-Sensor. ?Sensor ont:BEMOP-hasLocation ?Room. ?Room rdf:type ont:BI-Room. ?SensorOutput ont:SSNOP-isProduceBy ?Sensor. ?SensorOutput ont:SSNOP-hasObservation ?Observation. ?Observation ont:SSNOP-hasObservationValue ?ObservationValue. ?ObservationValue ont:BEMOP-hasEnvironmentObjectType ?A. ?A rdf:type ont:LightEnvironment. ?ObservationValue ont:hasValue ?Light. filter (?Light < 300) }</pre>		
Sensor	Room	Light
Sensor9	Room6	"200"^^<http://www.w3.org/2001/XMLSchema#integer>
Sensor8	Room5	"270"^^<http://www.w3.org/2001/XMLSchema#integer>

Fig. 17. Query result of rule D.

SPARQL query:		
<pre> PREFIX ont: <http://www.semanticweb.org/gan/ontologies/2016/11/untitled-ontology-1#> PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#> SELECT ?Room ?AreaMeasure ?AirPollutantSensorQuantity WHERE { ?Room rdf:type ont:BI-Room. ?Room ont:BIOP-hasAreaMeasure ?A. ?A ont:hasValue ?AreaMeasure. ?Room ont:BIOP-hasAirPollutantSensorQuantity ?B. ?B ont:hasValue ?AirPollutantSensorQuantity. filter((?AreaMeasure < 50 && ?AirPollutantSensorQuantity < 1) (?AreaMeasure >= 50 && ?AreaMeasure < 100 && ?AirPollutantSensorQuantity < 2) (?AreaMeasure >= 100 && ?AreaMeasure < 500 && ?AirPollutantSensorQuantity < 3) (?AreaMeasure >= 500 && ?AreaMeasure < 1000 && ?AirPollutantSensorQuantity < 5) (?AreaMeasure >= 1000 && ?AreaMeasure < 3000 && ?AirPollutantSensorQuantity < 6) (?AreaMeasure >= 3000 && ?AirPollutantSensorQuantity < 3 * ?AreaMeasure / 1000)) }</pre>		
Room	AreaMeasure	AirPollutantSensorQuantity
Room3	"53.72"^^<http://www.w3.org/2001/XMLSchema#float>	"1"^^<http://www.w3.org/2001/XMLSchema#integer>
Room6	"110.00"^^<http://www.w3.org/2001/XMLSchema#float>	"2"^^<http://www.w3.org/2001/XMLSchema#integer>

Fig. 18. Query result of rule E.

Table 6

Environmental checking results of each room.

Room No. \Environment Indicators	Radon	Carbon monoxide	Temperature	Light	Correlation between the quantity of air pollutant sensors and room area
Room 1	Pass	Pass	Pass	/	Pass
Room 2	Warning	Warning	Warning	/	Pass
Room 3	Warning	Warning	Warning	/	Warning
Room 4	/	/	/	Pass	Pass
Room 5	/	/	/	Warning	Pass
Room 6	/	/	/	Warning	Warning

The proposed framework is validated only on limited given instances of a real case, as presented in the above section. The other building environmental information can be modeled and reasoned similarly, and the same result can be got, since the ontology-model can be readily extended or modified according to the specific requirements as the ontological modeling approach is flexible. Although, only the specific regulatory constraints for the case study are illustrated in this paper, we suggest that constraints from other regulations can also be represented, for other objects, as long as they belong to the constraint types identified in this research. Further research should use a large number of examples to validate the generality of this framework.

5. Conclusion and limitation

With the usage of sensors for the building environmental monitoring, the volume of heterogeneous data is collected for monitoring and checking. During this process, the sharing between different stakeholders and interoperability between different information systems still remain challenging.

To fill this gap and achieve effective representation, management of such heterogeneous data and further enable the automated compliance checking, an ontology-based framework for building environmental monitoring and compliance checking under BIM environment is proposed with using ontology and semantic web technologies.

This framework integrates building information from BIM, environmental information provided by sensors, and regulatory information based on building regulations and design requirements, and enables the semantic representation and inference of the information required for building environmental monitoring and compliance checking, which takes a step towards filling this gap.

To develop this framework, various technical issues are addressed in this study: 1) Development of four specific ontologies for representing and integrating various information to support the building environmental monitoring and compliance checking; 2) Information extraction of sensors from BIM models; 3) Conversion of information from BIM and sensors to the RDF data model; 4) Compliance checking and search of the substandard instances in the ontology by using SPARQL rules.

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Table 7
Comparison results.

Rule No.	Constraint topics	Constraint content	Results of manual judgment (Warning)	Results of automatic reasoning (Warning)	Comparison between the results of automatic reasoning and manual judgment
A	Radon	The concentration of radon should be less than or equal to 400 Bq/m ³	Sensor 3, 5	Sensor 3, 5	Consistent
B	Carbon monoxide	The concentration of carbon monoxide should be less than 20 ppm	Sensor 3, 5	Sensor 3, 5	Consistent
C	Temperature	The temperature range should be between 0 and 38° Celsius	Sensor 4, 6	Sensor 4, 6	Consistent
D	Light	The light value should be greater than or equal to 300 lx	Sensor 8, 9	Sensor 8, 9	Consistent
E	Correlation between the quantity of air pollutant sensors and room area	When the room area is less than 50 m ² , the quantity of air pollutant sensors is greater than or equal to 1; When the room area is greater than or equal to 50 m ² and less than 100 m ² , the quantity of air pollutant sensors is greater than or equal to 2; When the room area is greater than or equal to 100 m ² and less than 500 m ² , the quantity of air pollutant sensors is greater than or equal to 3; When the room area is greater than or equal to 500 m ² and less than 1000 m ² , the quantity of air pollutant sensors is greater than or equal to 5; When the room area is greater than or equal to 1000 m ² and less than 3000 m ² , the quantity of air pollutant sensors is greater than or equal to 6; When the room area is greater than or equal to 3000 m ² , the quantity of air pollutant sensors is greater than or equal to 3/1000m ² ;	Room 3, 6	Room 3, 6	Consistent

station project is used as a testing case. The testing results validate the feasibility and effectiveness of the proposed framework. The proposed framework would be highly beneficial to promote the effective inter-operation of monitoring systems and data sharing between stakeholders. It is also helpful the domain knowledge reuse.

However, we recognize that our approach has following limitations.

- 1) This framework assumes that the BIM model is semantically rich, and the building information of the project is integrated into BIM model and can be extracted from it.
- 2) The framework is based on several standard ontologies. In this work, only the simple prototype ontologies are developed. However, the ontology development is time-consuming, considerable initial work needs to be done. In the near future, the development of these ontologies can also serve as the starting point for further research.
- 3) Aside from the building interior environment, the external environment, which includes the geographical environment, climate environment, and spatial environment of buildings, should also be monitored in the future. In addition, GIS and the external environment can be integrated and considered for building environmental monitoring and compliance checking.

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References

- [1] Y. Lu, Z. Wu, R. Chang, Y. Li, Building Information Modeling (BIM) for green buildings: a critical review and future directions, *Autom. Construct.* 83 (2017) 134–148.
- [2] M. Dibley, H. Li, Y. Rezgui, J. Miles, An ontology framework for intelligent sensor-based building monitoring, *Autom. Construct.* 28 (12) (2012) 1–14.
- [3] J.D. Goedert, P. Meadati, Integrating construction process documentation into building information modeling, *J. Construct. Eng. Manag.* 134 (7) (2015) 509–516.
- [4] E.R.P. Farr, P.A.E. Piroozfar, D. Robinson, BIM as a generic configurator for facilitation of customisation in the AEC industry, *Autom. Construct.* 45 (45) (2014) 119–125.
- [5] S. Zhang, J. Teizer, J.K. Lee, C.M. Eastman, M. Venugopal, Building information modeling (BIM) and safety: automatic safety checking of construction models and schedules, *Autom. Construct.* 29 (4) (2013) 183–195.
- [6] P. Meadati, L. Ap, J. Irizarry, BIM – a knowledge repository, *Proceedings of the 46th Annual International Conference of the Associated Schools of Construction*, vol 12, 2010.
- [7] P.B. Pishdad, Y.J. Beliveau, Integrating multi-party contracting risk management (MPCRM) model with building information modeling (BIM), *CIB W78 27th International Conference on Applications of IT in the AEC Industry*, 2010, pp. 16–19.
- [8] E.M. Wetzel, W.Y. Thabet, The use of a BIM-based framework to support safe facility management processes, *Autom. Construct.* 60 (2015) 12–24.
- [9] T.W. Kang, C.H. Hong, A study on software architecture for effective BIM/GIS-based facility management data integration, *Autom. Construct.* 54 (2015) 25–38.
- [10] Y.C. Lin, Y.C. Su, Developing mobile- and BIM-based integrated visual facility maintenance management system, *Sci. World J.* 2013 (7) (2013).
- [11] M. Arslan, Z. Riaz, A.K. Kiani, S. Azhar, Real-time environmental monitoring, visualization and notification system for construction H&S management, *Electron. J. Inf. Technol. Construct.* 19 (11) (2014) 72–91.
- [12] Z. Riaz, D.J. Edwards, E.A. Parn, C. Shen, F. Pena-Mora, BIM and sensor-based data management system for construction safety monitoring, *J. Eng. Des. Technol.* 15 (6) (2017) 738–753.
- [13] W. Natephra, A. Motamedi, N. Yabuki, T. Fukuda, Integrating 4D thermal information with BIM for building envelope thermal performance analysis and thermal comfort evaluation in naturally ventilated environments, *Build. Environ.* 124 (2017) (2017) 194–208.
- [14] K. Suprabhas, H.N. Dib, Integration of BIM and utility sensor data for facilities management, *Asce International Workshop on Computing in Civil Engineering*, vol 2016, 2017, pp. 26–33.
- [15] T.R. Gruber, Toward principles for the design of ontologies used for knowledge sharing? *Int. J. Hum. Comput. Stud.* 43 (5–6) (1993) 907–928.
- [16] D. Richards, S.J. Simoff, Design ontology in context — a situated cognition approach to conceptual modelling, *Artif. Intell. Eng.* 15 (2) (2001) 121–136.
- [17] K.B. Sørensen, P. Christiansson, K. Svidt, Ontologies to support RFID-based link between virtual models and construction components, *Comput. Aided Civ. Infrastruct. Eng.* 25 (25) (2010) 285–302.
- [18] E. Corry, P. Pauwels, S. Hu, M. Keane, J. O'Donnell, A performance assessment ontology for the environmental and energy management of buildings, *Autom. Construct.* 57 (2015) 249–259.
- [19] P. Pauwels, R.D. Meyer, J.V. Campenhout, Interoperability for the design and construction industry through semantic web technology, *International Conference on Semantic and Digital Media Technologies*, 2010, pp. 143–158.
- [20] E.P. Karan, J. Irizarry, J. Haymaker, BIM and GIS integration and interoperability based on semantic web technology, *J. Comput. Civ. Eng.* 30 (3) (2015) 4015043.
- [21] C. Mignard, C. Nicolle, Merging BIM and GIS using ontologies application to urban facility management in ACTIVE3D, *Comput. Ind.* 65 (9) (2014) 1276–1290.
- [22] E.W. Patton, P. Seyed, P. Wang, L. Fu, F.J. Dein, R.S. Bristol, D.L. McGuinness, SemantEco: a semantically powered modular architecture for integrating

- distributed environmental and ecological data, *Future Generat. Comput. Syst.* 36 (3) (2014) 430–440.
- [23] C. Métral, G. Falquet, K. Karatzas, Ontologies for the integration of air quality models and 3D city models, *Proceedings of 2nd Workshop of COST Action C21-townology: Ontologies for Urban Development: Conceptual Models for Practitioners*, 2007, pp. 18–33.
- [24] M.M. Oprea, *AIR_POLLUTION_Onto: an ontology for air pollution analysis and control*, *IFIP Adv. Inf. Commun. Technol.* 296 (2009) 135–143.
- [25] J. Adeleke, D. Moodley, An ontology for proactive indoor environmental quality monitoring and control, *Proceedings of the 2015 Annual Research Conference on South African Institute of Computer Scientists and Information Technologists. ACM*, 2015, p. 2.
- [26] M. Stocker, M. Rönkkö, M. Kolehmainen, Towards an ontology for situation assessment in environmental monitoring, *International Congress on Environmental Modelling and Software*, 2014, pp. 1280–1288.
- [27] H. Pundt, Y. Bishr, Domain ontologies for data sharing-an example from environmental monitoring using field GIS, *Comput. Geosci.* 28 (1) (2002) 95–102.
- [28] W. Solihin, C. Eastman, Classification of rules for automated BIM rule checking development, *Autom. Construct.* 53 (2015) 69–82.
- [29] S.J. Fenves, Tabular decision logic for structural design, *Am. Soc. Civ. Eng.* 92 (6) (1966) 473–490.
- [30] N.O. Nawari, Automating codes conformance in structural domain, *International Workshop on Computing in Civil Engineering*, 2011, pp. 569–577.
- [31] J. Choi, J. Choi, I. Kim, Development of BIM-based evacuation regulation checking system for high-rise and complex buildings, *Autom. Construct.* 46 (10) (2014) 38–49.
- [32] P. Pauwels, D.V. Deursen, R. Verstraeten, J.D. Roo, R.D. Meyer, R.V.D. Walle, J.V. Campenhout, A semantic rule checking environment for building performance checking, *Autom. Construct.* 20 (5) (2011) 506–518.
- [33] B.T. Zhong, L.Y. Ding, H.B. Luo, Y. Zhou, Y.Z. Hu, H.M. Hu, Ontology-based semantic modeling of regulation constraint for automated construction quality compliance checking, *Autom. Construct.* 28 (12) (2012) 58–70.
- [34] D.Y. Lee, H.L. Chi, J. Wang, X. Wang, C.S. Park, A linked data system framework for sharing construction defect information using ontologies and BIM environments, *Autom. Construct.* 68 (2016) 102–113.
- [35] P. Pauwels, T.M.D. Farias, C. Zhang, A. Roxin, J. Beetz, J.D. Roo, C. Nicolle, A performance benchmark over semantic rule checking approaches in construction industry, *Adv. Eng. Inf.* 33 (2017) 68–88.
- [36] S.M. İlal, H.M. Günaydin, Computer representation of building codes for automated compliance checking, *Autom. Construct.* 82 (2017) 43–58.
- [37] M. Tao, K. Ota, M. Dong, Ontology-based data semantic management and application in IoT- and cloud-enabled smart homes, *Future Generat. Comput. Syst.* 76 (2017) 528–539.
- [38] D.T.J. O Sullivan, M.M. Keane, D. Kelliher, R.J. Hitchcock, Improving building operation by tracking performance metrics throughout the building lifecycle (BLC), *Energy Build.* 36 (11) (2004) 1075–1090.
- [39] J. Beetz, J.V. Leeuwen, B.D. Vries, IfcOWL: a case of transforming EXPRESS schemas into ontologies, *AI EDAM (Artif. Intell. Eng. Des. Anal. Manuf.)* 23 (23) (2009) 89–101.
- [40] W.C.S.S. Group, *Semantic Sensor Network Ontology*, vol 2017, (2009) p. W3C.
- [41] L. Zhang, R.R.A. Issa, Ontology-based partial building information model extraction, *J. Comput. Civ. Eng.* 27 (6) (2013) 576–584.
- [42] M.P. Nepal, S. Staub-French, R. Pottinger, A. Webster, Querying a building information model for construction-specific spatial information, *Adv. Eng. Inf.* 26 (4) (2012) 904–923.
- [43] M.P. Nepal, S. Staub-French, R. Pottinger, J. Zhang, Ontology-based feature modeling for construction information extraction from a building information model, *J. Comput. Civ. Eng.* 27 (5) (2013) 555–569.
- [44] S. Demir, J.H. Garrett, B. Akinci, O. Akin, A semantic web-based approach for representing and reasoning with vocabulary for computer-based standards processing, *The Proceedings of the 2010 International Conference on Computing in Civil and Building Engineering (ICCCBE'10)*, Nottingham, UK, 2010.
- [45] K. Ahmed, *dotNetRDF, Vol. 2017, 2016, P. An Open Source.NET Library for RDF*.
- [46] B.T. Zhong, H.B. Luo, Y.Z. Hu, J. Sun, Ontology-based approach for automated quality compliance checking against regulation in metro construction project, *Proceedings of the 1st International Workshop on High-speed and Intercity Railways*, 2012, pp. 385–396.
- [47] E. Prud'Hommeaux, A. Seaborne, *SPARQL Query Language for RDF, W3C Recommendation*, vol 2016, (2008).
- [48] C. Eastman, J.M. Lee, Y.S. Jeong, J.K. Lee, Automatic rule-based checking of building designs, *Autom. Construct.* 18 (8) (2009) 1011–1033.
- [49] *Code for Indoor Environmental Pollution Control of Civil Building Engineering*, GB 50325-2010, China Building Industry Press, Beijing, 2013.
- [50] *Emission Standard of Environment Noise for Boundary of Construction Site*, GB 12523-2011, China Building Industry Press, Beijing, 2011.
- [51] *Indoor Air Quality Standard*, GB/T 18883-2002, China Building Industry Press, Beijing, 2002.
- [52] *Standard for Lighting Design of Buildings*, GB 50034-2013, China Building Industry Press, Beijing, 2013.
- [53] Burcu Akinci, Situational awareness in construction and facility management, *Front. Eng. Manag.* 1 (3) (2014) 283–289.